

The Game of Nim, Bouton's Theorem, and Fibonacci Bit-Patterns

A Complete Mathematical Derivation of Project Euler Problem 301

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Abstract

We present a rigorous, self-contained mathematical analysis and derivation of Project Euler Problem 301. By employing Bouton's classic theorem from Combinatorial Game Theory, we express the winning/losing positions of three-heap Nim via the bitwise XOR sum (Nim-sum). We analyze the specific family of configurations $(n, 2n, 3n)$ and establish an exact algebraic bridge between bitwise addition and standard arithmetic, reducing the condition $X(n, 2n, 3n) = 0$ to carryless addition $n \& 2n = 0$. We prove that this condition is equivalent to the absence of consecutive ones in the binary representation of n , and we enumerate such integers on the interval $[1, 2^{30}]$ via a combinatorial bijection with the Fibonacci sequence, arriving at the exact count $\mathcal{N}(2^{30}) = F_{32} = 2\,178\,309$.

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1 Introduction and Problem Formulation

1.1 Rules of Impartial Normal-Play Nim

Nim is a classic impartial game played by two players with a finite collection of heaps containing discrete tokens (or stones). The game operates under the **normal play convention**:

- (i) Players alternate turns.
- (ii) On each turn, the active player chooses a single non-empty heap and removes any positive integer number of stones from it (up to the entire heap).
- (iii) The game ends when no stones remain in any heap.
- (iv) The first player unable to make a valid move (because all heaps are empty) loses; equivalently, the player who makes the final move wins.

A game state with three heaps is denoted by the triple $(n_1, n_2, n_3) \in \mathbb{N}_0^3$, where n_i represents the number of stones in the i -th heap.

1.2 Combinatorial Game States: $\mathcal{P}$ -Positions and $\mathcal{N}$ -Positions

In Combinatorial Game Theory (CGT), every position in a finite impartial game without cycles belongs to one of two mutually exclusive classes:

- **$\mathcal{P}$ -positions (Previous-player winning / Losing for the next player):** Any valid move from a state in $\mathcal{P}$ leads to a state in $\mathcal{N}$. The terminal state $(0, 0, 0)$ is in $\mathcal{P}$.
- **$\mathcal{N}$ -positions (Next-player winning / Winning for the next player):** There exists at least one legal move from any state in $\mathcal{N}$ to some state in $\mathcal{P}$.

The characteristic evaluation function $X(n_1, n_2, n_3)$ is defined as:

$$X(n_1, n_2, n_3) = \begin{cases} 0, & \text{if } (n_1, n_2, n_3) \in \mathcal{P} \quad (\text{second player can force a win}), \\ \neq 0, & \text{if } (n_1, n_2, n_3) \in \mathcal{N} \quad (\text{first player can force a win}). \end{cases} \quad (1.1)$$

1.3 Example: The Winning Strategy from Starting Position $(1, 2, 3)$

To illustrate the mechanism of $\mathcal{P}$ -positions and the winning strategy, consider the starting state $(1, 2, 3)$. We will show that $(1, 2, 3) \in \mathcal{P}$, meaning that the second player can always force a win regardless of the first player's initial move.

The first player has six possible opening moves from $(1, 2, 3)$:

1. **Reduce Heap 1** ($1 \rightarrow 0$): The state becomes $(0, 2, 3)$.

Winning Response by Player 2: Player 2 reduces Heap 3 from $3 \rightarrow 2$, leaving $(0, 2, 2)$. This is a two-heap symmetric position. Whatever Player 1 takes from one heap of size 2, Player 2 mirrors by taking the exact same amount from the other heap, guaranteeing the last move.

2. **Reduce Heap 2** ($2 \rightarrow 1$): The state becomes $(1, 1, 3)$.

Winning Response by Player 2: Player 2 removes all stones from Heap 3 ($3 \rightarrow 0$), leaving $(1, 1, 0)$. Again, Player 2 adopts the mirror strategy on the two equal heaps of size 1.

3. **Reduce Heap 2** ($2 \rightarrow 0$): The state becomes $(1, 0, 3)$.

Winning Response by Player 2: Player 2 reduces Heap 3 from $3 \rightarrow 1$, leaving $(1, 0, 1)$, securing the symmetric mirror state.

4. **Reduce Heap 3** ($3 \rightarrow 2$): The state becomes $(1, 2, 2)$.

Winning Response by Player 2: Player 2 takes the single stone from Heap 1 ($1 \rightarrow 0$), leaving $(0, 2, 2)$.

5. **Reduce Heap 3** ($3 \rightarrow 1$): The state becomes $(1, 2, 1)$.

Winning Response by Player 2: Player 2 removes all stones from Heap 2 ($2 \rightarrow 0$), leaving $(1, 0, 1)$.

6. **Reduce Heap 3** ($3 \rightarrow 0$): The state becomes $(1, 2, 0)$.

Winning Response by Player 2: Player 2 reduces Heap 2 from $2 \rightarrow 1$, leaving $(1, 1, 0)$.

In every single branch, Player 2 transitions the game to a state of two equal non-empty heaps $(k, k, 0)$, from which any move by Player 1 allows Player 2 to maintain equality or win immediately. An example sequence of optimal play is:

Player 1 moves:	$(1, 2, 3) \longrightarrow (1, 2, 1)$
Player 2 responds:	$(1, 2, 1) \longrightarrow (1, 0, 1)$
Player 1 moves:	$(1, 0, 1) \longrightarrow (0, 0, 1)$
Player 2 responds:	$(0, 0, 1) \longrightarrow (0, 0, 0)$ (Player 2 wins!)

Hence, $X(1, 2, 3) = 0$.

1.4 The Target Question

Project Euler Problem 301 asks:

$$\text{For how many positive integers } n \leq 2^{30} \text{ does } X(n, 2n, 3n) = 0? \quad (1.2)$$

2 Bouton's Theorem and the Nim-Sum

2.1 Definition: The Nim-Sum

Definition 2.1 (Nim-Sum). Let $a, b \in \mathbb{N}_0$ have binary expansions:

$$a = \sum_{i=0}^{\infty} a_i 2^i, \quad b = \sum_{i=0}^{\infty} b_i 2^i \quad (a_i, b_i \in \{0, 1\}). \quad (2.1)$$

The **Nim-sum** of a and b , denoted $a \oplus b$, is defined by bitwise addition modulo 2:

$$a \oplus b = \sum_{i=0}^{\infty} (a_i \oplus b_i) 2^i, \quad \text{where } a_i \oplus b_i = (a_i + b_i) \bmod 2. \quad (2.2)$$

The algebraic structure $(\mathbb{N}_0, \oplus)$ is an infinite elementary abelian 2-group:

- (i) **Associativity and Commutativity:** $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ and $a \oplus b = b \oplus a$.
- (ii) **Identity Element:** $a \oplus 0 = a$ for all $a \in \mathbb{N}_0$.
- (iii) **Self-Inversion:** $a \oplus a = 0$ for all $a \in \mathbb{N}_0$.

2.2 Bouton's Theorem

Theorem 2.2 (Bouton, 1901). *In an m -heap game of Nim with heap sizes $(h_1, h_2, \dots, h_m) \in \mathbb{N}_0^m$, a position is a $\mathcal{P}$ -position (losing for the player whose turn it is) if and only if the Nim-sum of all heap sizes is zero:*

$$\bigoplus_{j=1}^m h_j = 0. \quad (2.3)$$

Equivalently, $X(h_1, h_2, \dots, h_m) = \bigoplus_{j=1}^m h_j$.

Proof. Let $\mathcal{Z} = \{(h_1, \dots, h_m) \in \mathbb{N}_0^m : \bigoplus_{j=1}^m h_j = 0\}$. We prove by complete induction on the total stone count $S = \sum_{j=1}^m h_j$ that $\mathcal{Z}$ satisfies the axiomatic properties of $\mathcal{P}$ -positions:

1. **Base Case** ($S = 0$): The only state is $(0, \dots, 0)$. Its Nim-sum is $0 \oplus \dots \oplus 0 = 0 \in \mathcal{Z}$. The active player has no moves and loses, so $(0, \dots, 0) \in \mathcal{P}$.
2. **Every move from $\mathcal{Z}$ leads outside $\mathcal{Z}$** ($\mathcal{P} \rightarrow \mathcal{N}$): Let $(h_1, \dots, h_m) \in \mathcal{Z}$, so $\bigoplus_{j=1}^m h_j = 0$. A legal move strictly decreases one heap, say $h_k \rightarrow h'_k$ with $0 \leq h'_k < h_k$. The new Nim-sum V' is:

$$V' = \left(\bigoplus_{j \neq k} h_j \right) \oplus h'_k = \left(\bigoplus_{j=1}^m h_j \right) \oplus h_k \oplus h'_k = 0 \oplus h_k \oplus h'_k = h_k \oplus h'_k. \quad (2.4)$$

Since $h'_k \neq h_k$, $h_k \oplus h'_k \neq 0$. Thus, $V' \neq 0$, meaning the new state is not in $\mathcal{Z}$.

3. **From any state not in $\mathcal{Z}$, there exists a move into $\mathcal{Z}$** ($\mathcal{N} \rightarrow \mathcal{P}$): Suppose $(h_1, \dots, h_m) \notin \mathcal{Z}$, so $V = \bigoplus_{j=1}^m h_j \neq 0$. Let $d = \lfloor \log_2 V \rfloor$ denote the most significant bit (MSB) of V , so bit d of V is 1. Since bit d of V is the parity of the d -th bits of all h_j , there exists at least one heap h_k whose d -th bit is 1.

We choose the replacement size $h'_k = h_k \oplus V$.

- For all bit indices $i > d$, the i -th bit of V is 0, so bit i of h'_k matches bit i of h_k .
- At bit index d , bit d of h_k is 1 and bit d of V is 1, so bit d of h'_k is $1 \oplus 1 = 0$.

Consequently, $h'_k < h_k$, so reducing heap k to h'_k is a valid Nim move. The new Nim-sum is:

$$V' = V \oplus h_k \oplus h'_k = V \oplus h_k \oplus (h_k \oplus V) = 0. \quad (2.5)$$

Thus the resulting state is in $\mathcal{Z}$.

By mathematical induction, $\mathcal{Z} = \mathcal{P}$, completing the proof. $\square$

3 Algebraic Reduction of $X(n, 2n, 3n) = 0$

3.1 Condition Equivalence

For the configuration $(n, 2n, 3n)$, Theorem 2.2 states:

$$X(n, 2n, 3n) = 0 \iff n \oplus 2n \oplus 3n = 0. \quad (3.1)$$

Using the self-inverting group property of $\oplus$:

$$n \oplus 2n \oplus 3n = 0 \iff n \oplus 2n = 3n. \quad (3.2)$$

Observing that $3n = n + 2n$ in ordinary arithmetic, this becomes:

$$n \oplus 2n = n + 2n. \quad (3.3)$$

3.2 Arithmetic Addition vs. Bitwise XOR

Lemma 3.1 (Arithmetic and Bitwise Addition Identity). *For any $a, b \in \mathbb{N}_0$,*

$$a + b = (a \oplus b) + 2(a \& b), \quad (3.4)$$

where $\&$ denotes the bitwise AND operator.

Proof. Let $a = \sum_{i=0}^{\infty} a_i 2^i$ and $b = \sum_{i=0}^{\infty} b_i 2^i$ be the binary representations of a and b , with $a_i, b_i \in \{0, 1\}$. For each single bit position $i \geq 0$, we test all four possible binary assignments for (a_i, b_i) :

a_i	b_i	$a_i + b_i$	$a_i \oplus b_i$	$a_i \& b_i$	$(a_i \oplus b_i) + 2(a_i \& b_i)$
0	0	0	0	0	$0 + 2(0) = 0$
0	1	1	1	0	$1 + 2(0) = 1$
1	0	1	1	0	$1 + 2(0) = 1$
1	1	2	0	1	$0 + 2(1) = 2$

In every case, $a_i + b_i = (a_i \oplus b_i) + 2(a_i \& b_i)$ holds identically. Multiplying by 2^i and summing over all $i \geq 0$:

$$\sum_{i=0}^{\infty} (a_i + b_i) 2^i = \sum_{i=0}^{\infty} (a_i \oplus b_i) 2^i + 2 \sum_{i=0}^{\infty} (a_i \& b_i) 2^i. \quad (3.5)$$

By linearity of summation, this yields $a + b = (a \oplus b) + 2(a \& b)$. $\square$

Corollary 3.2 (Carryless Addition Criterion). *For any non-negative integers $a, b \in \mathbb{N}_0$,*

$$a \oplus b = a + b \iff a \& b = 0. \quad (3.6)$$

Proof. From Lemma 3.1, $(a + b) - (a \oplus b) = 2(a \& b)$. Since $a \& b \geq 0$, the difference vanishes if and only if $a \& b = 0$. $\square$

Applying Corollary 3.2 to $a = n$ and $b = 2n$:

$$X(n, 2n, 3n) = 0 \iff n \& (2n) = 0. \quad (3.7)$$

4 The "No Consecutive Ones" Bitwise Characterization

4.1 Bit-Shift Analysis

Let $n \in \mathbb{N}$ have binary representation $n = \sum_{i=0}^{\infty} n_i 2^i$ with $n_i \in \{0, 1\}$. Multiplying n by 2 corresponds to an arithmetic left-shift by 1 bit:

$$2n = \sum_{i=0}^{\infty} n_i 2^{i+1} = \sum_{j=1}^{\infty} n_{j-1} 2^j. \quad (4.1)$$

Thus, the i -th bit of $2n$ is:

$$(2n)_0 = 0, \quad (2n)_i = n_{i-1} \quad \text{for all } i \geq 1. \quad (4.2)$$

4.2 Bitwise AND Evaluation

The i -th bit of the bitwise AND product $n \& (2n)$ is:

$$(n \& 2n)_0 = n_0 \cdot (2n)_0 = n_0 \cdot 0 = 0, \quad (4.3)$$

$$(n \& 2n)_i = n_i \cdot (2n)_i = n_i \cdot n_{i-1} \quad \text{for all } i \geq 1. \quad (4.4)$$

Therefore, $n \& (2n) = 0$ if and only if every bit of the product is 0:

$$n \& (2n) = 0 \iff n_i \cdot n_{i-1} = 0 \quad \text{for all } i \geq 1. \quad (4.5)$$

Theorem 4.1 (Bit-Pattern Characterization). *A positive integer n satisfies $X(n, 2n, 3n) = 0$ if and only if the binary expansion of n contains no adjacent ones (that is, n contains no substring **11** in base 2).*

5 Combinatorial Enumeration via Fibonacci Numbers

5.1 Binary Strings Without Consecutive Ones

Let $\mathcal{B}_k$ denote the set of binary words of length $k \geq 0$ over $\{0, 1\}$ containing no adjacent ones:

$$\mathcal{B}_k = \{w \in \{0, 1\}^k : w \text{ does not contain the substring } 11\}. \quad (5.1)$$

Let $a_k = |\mathcal{B}_k|$.

- **Base Cases:**

- $k = 0$: The empty word ε contains no **11** $\implies a_0 = 1$.
- $k = 1$: The words "0" and "1" both contain no **11** $\implies a_1 = 2$.

• **Recurrence for $k \geq 2$:** We partition $\mathcal{B}_k$ according to the prefix of each word $w \in \mathcal{B}_k$:

1. **Words starting with 0:** Any such word is of the form $0u$, where $u \in \mathcal{B}_{k-1}$. There are a_{k-1} such words.
2. **Words starting with 1:** Since **11** is forbidden, the second character must be 0. Any such word is of the form $10v$, where $v \in \mathcal{B}_{k-2}$. There are a_{k-2} such words.

Since these two subsets are disjoint and partition $\mathcal{B}_k$:

$$a_k = a_{k-1} + a_{k-2} \quad \text{for all } k \geq 2. \quad (5.2)$$

5.2 Connection to the Fibonacci Sequence

Recall the standard Fibonacci sequence $(F_m)_{m \geq 1}$:

$$F_1 = 1, \quad F_2 = 1, \quad F_m = F_{m-1} + F_{m-2} \quad (m \geq 3). \quad (5.3)$$

Lemma 5.1. *For every integer $k \geq 0$,*

$$a_k = F_{k+2}. \quad (5.4)$$

Proof. We proceed by induction on k :

- For $k = 0$: $a_0 = 1 = F_2 = F_{0+2}$.
- For $k = 1$: $a_1 = 2 = F_3 = F_{1+2}$.
- Inductive step: Assume $a_j = F_{j+2}$ holds for all $j < k$ ($k \geq 2$). Then:

$$a_k = a_{k-1} + a_{k-2} = F_{k+1} + F_k = F_{k+2}. \quad (5.5)$$

By mathematical induction, $a_k = F_{k+2}$ for all $k \geq 0$. □

6 Counting Solutions on $[1, 2^K]$

6.1 Integers in the Half-Open Interval $[0, 2^K)$

Every integer n in the range $0 \leq n < 2^K$ can be represented as a binary string of length exactly K (by prepending leading zeros if necessary). Since leading zeros do not introduce **11** substrings, there is a natural bijection between $\{n \in \mathbb{N}_0 : 0 \leq n < 2^K \text{ and } n \& 2n = 0\}$ and $\mathcal{B}_K$. By Lemma 5.1:

$$\#\{n \in \mathbb{N}_0 : 0 \leq n < 2^K \text{ and } n \& 2n = 0\} = a_K = F_{K+2}. \quad (6.1)$$

6.2 Exclusion of $n = 0$

The set above includes $n = 0$ ($0 \& 0 = 0$). Removing $n = 0$ leaves:

$$\#\{n \in \mathbb{N} : 1 \leq n < 2^K \text{ and } n \& 2n = 0\} = F_{K+2} - 1. \quad (6.2)$$

6.3 Inclusion of the Upper Endpoint $n = 2^K$

Consider the boundary integer $n = 2^K$:

- The binary expansion of 2^K is $1\underbrace{00\dots 00}_{K \text{ zeros}}$.
- This representation contains only a single 1-bit, and thus contains no adjacent ones.
- Directly checking: $2^K \& (2 \cdot 2^K) = 2^K \& 2^{K+1} = 0$.

Hence, $n = 2^K$ is always a valid solution.

6.4 Total Solution Count $\mathcal{N}(2^K)$

Summing the contributions:

$$\mathcal{N}(2^K) = (F_{K+2} - 1) + 1 = F_{K+2}. \quad (6.3)$$

Theorem 6.1 (Closed-Form Solution). *For any integer $K \geq 1$, the number of positive integers $n \leq 2^K$ such that $X(n, 2n, 3n) = 0$ is given by the $(K + 2)$ -th Fibonacci number:*

$$\mathcal{N}(2^K) = F_{K+2}. \quad (6.4)$$

7 Exact Numerical Evaluation for $K = 30$

Setting $K = 30$ in Theorem 6.1, the total number of solutions is:

$$\mathcal{N}(2^{30}) = F_{30+2} = F_{32}. \quad (7.1)$$

7.1 Table of Fibonacci Numbers (F_1 to F_{32})

m	F_m	m	F_m	m	F_m	m	F_m
1	1	9	34	17	1 597	25	75 025
2	1	10	55	18	2 584	26	121 393
3	2	11	89	19	4 181	27	196 418
4	3	12	144	20	6 765	28	317 811
5	5	13	233	21	10 946	29	514 229
6	8	14	377	22	17 711	30	832 040
7	13	15	610	23	28 657	31	1 346 269
8	21	16	987	24	46 368	32	2 178 309

7.2 Final Conclusion

The number of positive integers $n \leq 2^{30}$ for which $X(n, 2n, 3n) = 0$ is:

2 178 309